



THE COPPERBELT UNIVERSITY
MCS SCHOOL OF MEDICINE
END OF TERM ONE TEST – APRIL 2023
MBS 210 - PHYSIOLOGY

STUDENT NUM: 3102 **PROGRAMME:** MBChB

TIME ALLOWED: 2 hours

TOTAL MARK: 100

INSTRUCTIONS:

1. Do not write your name, phone number or anything that discloses your identity on any other page apart from this page. Defaulters will have their evaluation nullified.
2. Write your computer number on every other page
3. Answer ALL questions in all **SECTIONS (A, B & C)**
4. Answer **SECTION A** in the grid sheet provided

61.5

SECTION A ANSWER SHEET

COURSE CODE... MBS 210

COMPUTER NUMBER... 2

DATE... 20/04/2023

No	A	B	C	D	E
1			X		
2	X			X	
3					X
4		X			
5			X		
6		X			
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No	A	B	C	D	E
26	X				
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26

SECTION A: CHOOSE THE SINGLE BEST OPTION [40 marks]

EACH QUESTION CARRIES 1 MARK

1. In which phase of the cardiac muscle action potential, is the calcium ion permeability the highest?
 - A. Depolarization
 - B. Repolarization
 - ☒ C. Plateau
 - D. Hyperpolarization
 - E. Uniform all over
2. Which of the following best explains how sympathetic stimulation affects the heart?
 - A. Permeability of the S-A node to sodium decreases ✓
 - B. Permeability of the A-V node to sodium decreases
 - C. Permeability of the S-A node to potassium increases
 - ☒ D. There is an increased rate of upward drift of the resting membrane potential of the S-A node
 - E. Permeability of the cardiac muscle to calcium decreases
3. Myocardial contractility is best controlled with the secretions and action of _____.
 - ☒ A. Na^+ ✓
 - B. Sympathetic nerves
 - C. Parasympathetic nerves
 - D. Both B & C
 - E. None
4. Example for autoimmune disease in muscle?
 - A. Muscular dystrophy .
 - ☒ B. Myasthenia gravis ✓
 - C. Polio
 - D. Paralysis
 - E. Paresis
5. Reason for action potential to get faster conductance rate ?
 - A. Oligodendrocytes
 - B. Calmodulin
 - ☒ C. Nodes of ranvier ✓
 - D. Unmyelinated fibers
 - E. Graded potentials
6. Even with high sophisticated language technology, humans cannot control their own body smooth muscles at will. Which of the following enzymes is responsible for phosphorylating myosin light chains to activate smooth muscle contraction?
 - A. Calmodulin

- ☒ B. Myosin light chain kinase ✓
 - ☐ C. Myosin light chain phosphatase
 - ☐ D. Myosin ATPase
 - ☐ E. Actomyosin ATPase
7. Which characteristic or component is shared by skeletal muscle and smooth muscle?
- ☐ A. Thick and thin filaments arranged equally in sarcomeres
 - ☐ B. Troponin
 - ☒ C. Elevation of intracellular $[Ca^{2+}]$ for excitation-contraction coupling ✓
 - ☐ D. Spontaneous depolarization of the membrane potential
 - ☐ E. High degree of electrical coupling between cells
8. A 66-year-old woman with severe muscle weakness is hospitalized. The only abnormality in her laboratory values is an elevated serum K^+ concentration. In a muscle, the K^+ ions play a major role in _____.
- ☐ A. The resting membrane potential
 - ☒ B. Repolarization phase ✓
 - ☐ C. Are closed by depolarization
 - ☐ D. Are closed by repolarization
 - ☐ E. Are opened by depolarization
9. How does the single-unit smooth muscle differ from multiunit smooth muscles?
- ☐ A. Single-unit muscle contraction speed is slow, while multiunit is fast
 - ☐ B. Single-unit muscle has T-tubules, while multiunit does not
 - ☐ C. Single-unit muscles are not innervated by autonomic nerves
 - ☒ D. Single-unit muscle produces action potentials spontaneously that spreads to neighbouring cells, while multiunit does not ✓
 - ☐ E. They are all smooth muscles, they do not differ
10. In excitation-contraction of smooth muscle, calcium binds to which protein after influx into the cytoplasm?
- ☒ A. Calmodulin ✓
 - ☐ B. Tropomyosin
 - ☐ C. Troponin C
 - ☐ D. Troponin I
 - ☐ E. Myosin
11. While attending to an emergency care patient [soldier] who came from conflicted region of TIGRAY at Ethiopian border with Eritrea, which of the following are isotonic fluids used for IV infusion?
- ☐ A. 1.9% NaCl
 - ☒ B. 5% Dextrose ✓
 - ☐ C. Fingers Lactate
 - ☐ D. 9% NaCl

- E. 19% Dextrose solution
12. As a medical graduate, you got to infuse an IV solution at emergency department based on the particular patient's
- A. Liver-volume balance
 - B. Permanent residency status
 - ☒ C. Serum electrolyte values ✓
 - D. Attitude
 - E. Blood oxygen level
13. In cardiac muscles, both chambers contract at the same time during heart beats. The function of gap junction or nexus that are present between two cells in any tissue is _____.
- ☒ A. Transportation and communication between cells ✓
 - B. Cl^- ions follows Na^+ ions
 - C. H_2O follows NaCl
 - D. Ca^{++} pumped out
 - E. 3 K^+ pumped out
14. Example for integral proteins of cell membrane: _____
- A. Carriers
 - B. Channels
 - ☒ C. Integral protein band 3 ✓
 - D. Antigen
 - E. Spectrin
15. Plasma membrane function as semi permeable barrier aids to _____
- ☒ A. Maintains difference in composition of ICF & ECF & fluid in various organelles ✓
 - B. Protects cell from toxic substances
 - C. Excretion of waste products
 - D. Transport of nutrients
 - E. Cell growth
16. Example for Passive transport mechanism across cell membrane _____
- A. Voltage gated diffusion
 - ☒ B. Facilitated diffusion ✓
 - C. Exocytosis
 - D. Endocytosis
 - E. Sodium potassium pump
17. Example of an active transport mechanisms across cell membrane is _____
- A. Cytolysis
 - B. Facilitated diffusion
 - C. Filtration
 - ☒ D. Secondary active transport ✓
 - E. Osmosis

18. Skeletal muscle number is fixed at time of birth in humans. Which of the following hormones stimulate muscle hypertrophy?

- A. Estrogen
- ☒ B. Growth hormone ✓
- C. Oestrone
- D. Adrenaline
- E. T4

19. Functional significance of sarcoplasmic reticulum in muscle cells is _____

- A. Membranous sacs which encircles each myofibril
- ☒ B. Stores and release calcium ions (Ca^{++}) ✓
- C. Release of Na^+ triggers muscle contraction
- D. Develops action potential
- E. Releases sodium ions

20. Regarding contractile filaments in a skeletal muscle

- A. Function in the non contractile process
- B. Three types of filaments (Thick, medium and Thin)
- ☒ C. There are two thin filaments for every thick filament ✓
- D. Secrete Ca^{++}
- E. Myosin are thin filaments

* 21. Regarding Sarcomeres ?

- A. Compartments of arranged fibres
- ☒ B. Basic functional unit of a myofibril ✓
- C. Secrete Ca^{++}
- D. Generate action potential
- E. Has voltage

22. Regarding myosin filaments

- A. Thin filaments
- B. Functions as a motor protein which cannot achieve motion
- C. Convert ADP to glucose of motion
- ☒ D. Projections of each myosin molecule protrude outward (myosin head) ✓
- E. Actin has cross bridges

23. Regarding Actin filaments in skeletal muscles

- A. Thick filaments
- ☒ B. Actin molecules provide a site where a myosin head can attach ✓
- C. Tropomyosin and troponin are branched part of the thin filament
- D. Doesnot contract
- E. Has cross vridges

24. Physiotherapy is a branch of contemporary medicine that focuses on neuromuscular rehabilitation of patients. The total strength of a contraction in a muscle depends on the

- A. Motor neurons
- B. Less motor units
- ☒ C. Number of motor units that are activated ✓
- D. Stimulus strength
- E. Energy drink

25. LATCH phenomenon in a smooth muscle refers to:

- A. Na^+ is transported into the SR
- B. Myosin-binding sites are covered by tropomyosin
- ☒ C. Myosin heads remain attached to actin for long duration ✓
- D. Power stroke of myosin occurs
- E. Binding sites are open always

26. VISUALIZE an action potential curve, of a nerve. During depolarization phase ;

- ☒ A. It is the decrease in cell electronegativity inside ✓
- B. It is the increase in cell electropositivity outside
- C. It is the decrease in cell electropositivity inside of nerve membrane
- D. It is the decrease in cell size
- E. No voltage change

27. We are living in modern world of artificial intelligence with CHAT-GPT. AI robots with can now replace a human as an UNBIASED personal assistant. An action potential in a nerve fibre

- A. Has amplitude which varies directly with the strength of the stimulus
- B. Induces graded electrical currents in adjacent segments of the fibre
- C. Is associated with a transient decrease in membrane permeability to potassium ✓
- ☒ D. Is associated with a transient increase in membrane permeability to sodium ✓
- E. Is associated with a transient increase in membrane permeability to Aluminium

*28. About synapses;

- ☒ A. Electrical synapses can be found between a neuron and a neuron ✓
- B. Physical synapses can be found between a neuron and a bone
- C. Chemical synapses cannot be found between a neuron and a skin cell
- D. Chemical synapses can be found between a neuron and a bone
- E. Exist inside blood plasma

29. BMW group is at the forefront of global market in innovation of using AI sensors for automobiles. The following is related to nerve conduction senses in humans:

- A. Neurons not communicate with each other by synapses

- (B) A synapse is typically formed between an axon of a neuron and the dendrite of another neuron ✓
- C. Nerve impulses go from the axon to the nerve body
- D. The only form of transmission of an impulse from one neuron to another is direct electrical connection
- E. Nerve impulse is a fictional signal

30. Regarding the sodium potassium pump;

- (A) It transports sodium and potassium by breaking down ATP ✓
- B. Has no beta subunit with transportation occurring on the beta subunit
- C. 3 sites inside the cell are for potassium and 2 sites on the extracellular allow sodium to bind
- D. Pumps 2 Na⁺ outside to function
- E. Pumps chloride ions
- ATPase

SECTION B: STRUCTURED AND SHORT ANSWER QUESTIONS [50 marks]

Answer all the questions in the provided spaces within the question paper

1. State two functional differences between ENDOCYTOSIS and EXOCYTOSIS across the plasma membrane [4 marks]

- Endocytosis involves the internalization of extracellular material from the extracellular matrix into cells while exocytosis involves the removal of intracellular materials to the outside of the cell.

2 - Endocytosis is used for transport destruction of cell microbes and cell debris while exocytosis is use for distribution and transport of materials needed in other cells like hormones and enzymes.

2. What's your perspective on concept of ALL or NONE law? Give example [4 marks]

All or None law state that threshold frequency gives the maximum stimulus response for a muscle to contract. All stimulus below the threshold frequency does not produce muscle contraction to occurs.

- Immature heart beats is an example.

- If a nerve fiber responds at all to a stimulus, it responds completely by conducting an impulse.

- Greater intensity of stimulation triggers more impulses per second not stronger impulse

3. Explain the ionic basis for the self generation of voltage observed in the cardiac muscle [5 marks]

3
During rapid depolarization Na^+ ions influx occurs due to activation of Na^+ gated ion channels. During early repolarization sodium channel close and there is little efflux of Potassium ion in the muscle. During plateau there is influx of calcium ion that by making inside of the cell ^{positive} charged and outside positive charge, the potassium ion efflux occurs rapidly, thereby causing repolarization with inside cell negative and outside positive due to K^+ ions outside and Ca^{++} ions inside. The ionic concentration is restored during a resting membrane potential.

4. What is the CLINICAL significance of a long refractory period in cardiac muscle tissue? [3 marks]

1
2
Long refractory period in cardiac muscle prevents the summation of impulses and the occurrence of Tetanus. Tetanus is caused by sustained muscle contraction when there is a long refractory period tetanus can not occur. Long refractory period also enables the atria to empty themselves into ventricles for contraction of ventricles and pumping of blood outside of the heart.

5. With respect to the excitation-contraction coupling, give two differences between skeletal and smooth muscles [5 marks]

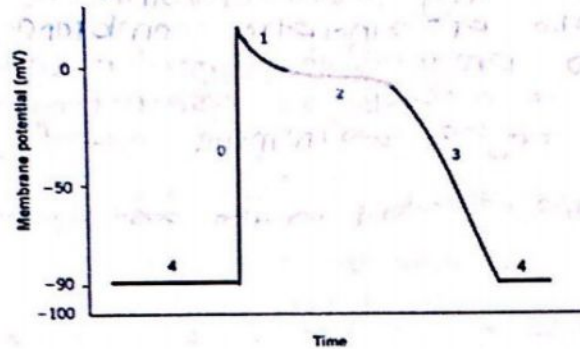
2
- In skeletal muscle depolarization is ^{caused} by the influx of sodium ion while in smooth muscle it is caused by the influx of calcium ion into the muscle cells.
- In skeletal muscle calcium binds to troponin while in smooth muscle ~~act~~ calcium binds to calmodulin.

6. IDENTIFY the types of nerve fibres in human body and CLASSIFY them [6 marks]

There are three types of nerve fibres: sensory neurone, interneurone and motor neurone. The nerve fibres can be classified according to structure and function. According to structure nerve fibres are unipolar,

~~Bt~~ Bipolar and multipolar. And according to functions nerve fibres have the sensory function, interconnecting function and motor functions. Sensory and motor neurones are multipolar while interneurons are unipolar. They both fall into the central nervous system and peripheral nervous system.

7. Label the parts 0 - 4 in the diagram and state the ionic basis for each [5 marks]



0. upstroke (rapid depolarization) influx of sodium ions into the cell caused by opening of sodium voltage gated channels.
 1. Early repolarization: closure of sodium voltage gated channel and little efflux of potassium ions cause by opening some K^+ slow channels.
 2. plateau: rapid influx of calcium ions caused by opening of most calcium slow long lasting channels and decrease in membrane K^+ permeability.
 3. Repolarization: rapid ion potassium efflux cause by open of most slow long lasting K^+ channels and increase permeability.
 4. Resting membrane potential: The concentration of K^+ and Ca^{2+} ions is restored by the sodium potassium pump. Inside the cell potassium goes in and calcium goes out.

8. American football game is adored in the nation of South Africa. Some of the veteran players have been diagnosed with muscular degenerative diseases in their middle age.

- a. What would happen if one of these players had increased potassium in the ECF. Outline the physiology that would be disrupted at plasma membrane level of a nerve. [3 marks]

Increased potassium in the ECF, hyperkalemia, decrease automaticity of the heart and inhibits it. It causes rapid hyperrepolarization which makes it longer for the production of an action potential. slow conduction of impulse by nerve and the nerve impulse will be weak.

b. Body muscles assist in maintaining the appropriate TONE and posture. Outline the derangements that occur in myasthenia-gravis in terms of NMJ in muscle physiology. [5 marks]

Chronic condition in which nerve impulses are not transmitted properly to the muscles

- Myasthenia gravis is due to lack of proper transmission of nerve impulses. The disease affects the NMJ by inhibiting nicotinic receptors found on the post synaptic membrane. The inhibition leads to reduction in synaptic vesicles that carry neurotransmitter to the synaptic knob at the pre synaptic membrane.
- It leads to progressive muscular weakness and paralysis.
- fatal when it affects the respiratory system.
- Etiology - maybe autoimmune affecting the myoneural junction.
- it affects the receptors on the post synaptic membrane.

c. What would happen if one of these players had Renshaw cell inhibition of neurons? Outline how the physiology of disruption occurs at the level of SYNAPSE. [5 marks]

- Neurons may also inhibit themselves in a negative feedback inhibition. A spinal neuron gives a collateral that synapses Renshaw cells which is inhibitory interneuron located in the anterior horn of the spinal cord. Renshaw cells in turn send back axons that inhibit the spinal motor neuron. These axons secrete an inhibitory transmitter that produces IPSP on cell bodies of motor neurons and inhibit them. Renshaw cell is an inhibitory cell excited by collaterals from an alpha motor neuron to project back and inhibit the same motor neurons.

9. Compare and contrast the concept of Osmolarity, Osmolality and Tonicity. Write about their clinical significance [5 marks]

Osmolarity is the measure of the osmoles of solutes per liter of a solution. It is affected by changes in volume caused by amount of solute, temperature and pressure.

Osmolality is the measure of the osmoles of solute per kg of a solvent. It is not affected by temperature and pressure.

Tonicity: this is the osmolality in relation to blood plasma. It can either be isotonic with equal balance of solute and solvent, hypotonic with more solvent than solute or hypertonic with more solute than solvent. All the three are used to determine blood solution administered into the body. Osmolality is used to measure body fluids. In a hypotonic solution cells will absorb water and in a hypertonic it will shrink. The tonicity of fluids must be known before administering fluids such as IV fluids into the body.

SECTION C: APPLIED SCENARIO QUESTION

[20 marks]

Answer all the questions in the spaces provided within the question paper

1. An 18-year-old person came to the emergency room after being stung by a bee while playing in the backyard when he suffered the sting, and within minutes he began having trouble in **BREATHING**. He had a "hive" rash over most of his body, along with increased difficulty breathing. When the emergency medical service (EMS) arrived, they administered epinephrine subcutaneously, which seemed to relieve most of the symptoms. In the emergency center, the boy was diagnosed with an anaphylactic reaction from the bee sting.

a) Describe the mechanism of excitation-contraction coupling in this muscle type

[5 marks]

3
There is increase in cytosolic calcium which comes from the extracellular matrix and the sarcoplasmic reticulum. Calcium binds to calmodulin found on thin filaments. An enzyme called light chain myosin kinase is activated. It causes the binding part of myosin light chain to be activated and properly oriented. Light chain myosin is phosphorylated. Myosin binds to actin and a cross bridge is formed. The cross bridge starts to rotate which initiates a power stroke. The sliding filament mechanism occurs and a myosin is excited and it contracts. Release of cytosolic calcium causes a drop in the detachment between the filaments. All the processes are reversed except phosphorylation.

2. A 35 years old gentleman Mr PP was having pain in his back. He is a veteran NATO soldier of Ukraine- Russia border. 2 years back, he had a surgery of hipbone injury. Now he is complaining of severe back muscle spasms and pain. At local clinic, on examination, his heart rate was 55 per minute, blood electrolytes were disturbed. All other data were normal.

Based on the above vignette, answer the following questions.

a) Discuss briefly the physiology [function] of anchoring proteins and regulating proteins at level of filaments in skeletal muscles. [5 M]

Anchoring proteins consists of Troponin and tropomyosin. Troponin provides a binding site for the calcium ion which causes change in conformation in actin filaments.

3 Troponin covers the myosin head binding site and determines the start and end of contraction. Binding of calcium to troponin causes the uncovering of the myosin binding site and leads to formation of cross bridges. The action of Troponin and troponin controls the start of contraction and they cover the binding site when contraction comes to an end.

- b) Discuss succinctly the types of IV fluids [with examples] that are needed to correct his electrolyte imbalance based on tonicity. Give REASONS and EXAMPLES [5 M]

- Saline

- half saline

- 5% Dextrose

3 IV fluids cause a balance in the electrolytes. Since it was a swelling, water molecules move out of a cell and accumulated in the interstitial space. An IV fluid which is hypertonic must be injected to absorb the accumulation in the swelling. Saline can be appropriate.

- c) The word BALANCE doesn't necessarily mean equality. Internal environment got to be in balance to stay HEALTHY. Elucidate the [malevolent] role of Donnan effect and how is it corrected back naturally, in restoring the cell size and body fluid electrolytes. Give EXAMPLE ions. [5 M]

3 Donnan effect is caused by accumulation of different charged substances on both side of the plasma membrane. This is because they can not diffuse through the membrane. It leads to an uneven distribution of charge across the cell membrane. It causes the inside of the cell osmotically active thereby causing cells to swell and burst due to movement of water molecules inside cells. The Donnan effect can be corrected by the sodium potassium pump. This pump brings $2K^+$ ions inside the cell and takes $3Na^+$ ions outside the cell. It can bring about balance in the charge across the cell membrane. It can also be reversed by a hyper hypertonic solution which will reverse the swelling of the cells.

THE END